

# OverTheWire Bandit Game

**Level 1:** The password for the next level is stored in a file called `readme` located in the home directory. Use this password to log into `bandit1` using SSH. Whenever you find a password for a level, use SSH (on port 2220) to log into that level and continue the game.

**Solution:** `ZjLjTmM6FvvyRnrb2rfNWOZOTa6ip5If` – `cat` command to read the file.

**Level 2:** The password for the next level is stored in a file called `-` located in the home directory

**Solution:** `263JGJPfgU6LtdEvgfWU1XP5yac29mFx` – `Cat ./-` directory just has a `'-'` no name

**Level 3:** The password for the next level is stored in a file called `--spaces in this filename-` located in the home directory.

**Solution -** `MNk8KNH3Usio41PRUEoDFPqfxLPIsmx` – When a file name starts with `-` you need to use the `CAT -` command.

**Level 4:** The password for the next level is stored in a hidden file in the `inhere` directory.

**Solution -** `2WmrDFRmJlq3IPxneAaMGhap0pFhF3NJ` – The file was hidden in a directory called `'inhere'` which was empty. To see hidden files you need to use the command `ls -a` and use `vim` to open the file.

**Level 5 -** The password for the next level is stored in the only human-readable file in the `inhere` directory. Tip: if your terminal is messed up, try the `"reset"` command.

**Solution -** `4oQYVPkxZOOEOO5pTW81FB8j8lxXGUQw` – in this level you need to go in to the directory `'inhere'` and `ls -l` which will list all the files. One by one I used the command `cat --file07` which is where the password was displayed in the only human-readable format.

**Level 6 –** The password for the next level is stored in a file somewhere under the `inhere` directory and has all of the following properties:

- human-readable
- 1033 bytes in size
- not executable

**Solution -** `HWasnPhtq9AVKe0dmk45nxy20cvUa6EG` – `cd` to `inhere` and use the command `find -type f -size 1033c` which will output `./maybe07/.file2`. After `cat ./maybe07/.file2` which will find the password.

**Level 7** – The password for the next level is stored **somewhere on the server** and has all of the following properties:

- owned by user bandit7
- owned by group bandit6
- 33 bytes in size

**Solution** - **morbNTDkSW6jIUc0ymOdMaLnOIFVAaj** – use the command `find / -type f -user bandit7 -group bandit 7 2>/dev/null` then cat the output

**Level 8** – The password for the next level is stored in the file **data.txt** next to the word **millionth**

**Solution** - **dfwvzFQi4mU0wfNbFOe9RoWskMLg7eEc** – since the word is in the same line as the password. We would use the `cat data.txt | grep millionth`. The cat command will display the output and using grep will display the password next to it.

**Level 9** – The password for the next level is stored in the file **data.txt** and is the only line of text that occurs only once

**Solution** - **4CKMh1JI91bUIZZPXqGanal4xvAg0JM** – in this situation we would use the `sort data.txt | uniq -u` command. This will sort the passwords and filter and find the password which appears once.

**Level 10** – The password for the next level is stored in the file **data.txt** in one of the few human-readable strings, preceded by several '=' characters.

**Solution** - **FGUW5ilLVJrxX9kMYMmlN4MgbpfMiqey** – we use the string command to distinguish human-readable strings. And use grep to filter the result and find the password as the password contains many equal signs before. The command would be like this: `strings data.txt | grep ===`

**Level 11** - The password for the next level is stored in the file **data.txt**, which contains base64 encoded data

**Solution** - **dtR173fZKb0RRsDFSGsg2RWnpNVj3qRr** – the solution is to use `base64 -d data.txt`. The command decodes the characters in data.txt and outputs the password.

**Level 12** – The password for the next level is stored in the file **data.txt**, where all lowercase (a-z) and uppercase (A-Z) letters have been rotated by 13 positions

**Solution** - **7x16WNeHli5YklhWsfFlqoognUTyj9Q4** – in this level I used **ROT13** command to sort the text in data.txt as it mentions on the tasks that the text has been rotated by 13 positions.

**Level 13** - The password for the next level is stored in the file **data.txt**, which is a hexdump of a file that has been repeatedly compressed. For this level it may be useful to create a

directory under /tmp in which you can work. Use mkdir with a hard to guess directory name. Or better, use the command “mktemp -d”. Then copy the datafile using cp, and rename it using mv (read the manpages!)

**Solution - FO5dwFsc0cbaliH0h8J2eUks2vdTDwAn** – In this task you need to use hex dumps to carry out extracting the file to retrieve the passwords. I first created a temp directory and copied the file there. You need to use the command hexdump -C -n 4 compressed\_data to view the file and see the first 4 characters which appears which will show you how to extract it. For example, bandit12@bandit:/tmp/tmp.W5t1vua6G9\$ cat hexdump\_data 00000000: 1f8b 0808 0650 b45e 0203 6461 7461 322e .....P.^..data2. – this shows how to extract the data and in this case, it is a gz. You will then do the following to extract the file: bandit12@bandit:/tmp/tmp.W5t1vua6G9\$ mv compressed\_data compressed\_data.gz bandit12@bandit:/tmp/tmp.W5t1vua6G9\$ ls compressed\_data.gz hexdump\_data bandit12@bandit:/tmp/tmp.W5t1vua6G9\$ gzip -d compressed\_data.gz. this process is then repeated until a file is printed with the password on the screen for the next level.

**Level 14** – The password for the next level is stored in /etc/bandit\_pass/bandit14 and can only be read by user bandit14. For this level, you don’t get the next password, but you get a private SSH key that can be used to log into the next level. Look at the commands that logged you into previous bandit levels and find out how to use the key for this level.

**Solution - in this task you need to use sshkey.private to ssh to the next level.**

**Level 15** - The password for the next level can be retrieved by submitting the password of the current level to port 30000 on localhost.

**Solution - 8xCjnmgoKbGLhHFAZIGE5Tmu4M2tKJQo** – in this task you need to cat connect to local host port 30000

**Level 16** - The password for the next level can be retrieved by submitting the password of the current level to port 30001 on localhost using SSL/TLS encryption.

**Solution - kSkvUpMQ7lBYyCM4GBPvCvT1BfWRy0Dx** – You need to use openssl s\_client - connect localhost:30001 then paste the Bandit15 password at the end of the certificate.

**Level 17** – The credentials for the next level can be retrieved by submitting the password of the current level to a port on localhost in the range 31000 to 32000. First find out which of these ports have a server listening on them. Then find out which of those speak SSL/TLS and which don’t. There is only 1 server that will give the next credentials, the others will simply send back to you whatever you send to it.

**Solution - x2gLTtjFwMOhQ8oWNbMN362QKxfRqGIO** – You first need to find what open ports there are by using `nmap -sV localhost -p 31000-32000`. This will then display open ports which you can use `openssl s_client -connect localhost:31790` to get the password.

**Level 18** - There are 2 files in the homedirectory: `passwords.old` and `passwords.new`. The password for the next level is in `passwords.new` and is the only line that has been changed between `passwords.old` and `passwords.new`

**Solution – cGWpMaKXVwDUNgPAVJbWYuGHVn9zl3j8** – You need to use the `diff` command to find the unique password for the next level by using `'diff passwords.old passwords.new'`

**Level 19** - The password for the next level is stored in a file `readme` in the homedirectory. Unfortunately, someone has modified `.bashrc` to log you out when you log in with SSH.

**Solution - 0qXahG8ZjOVMN9Ghs7iOWsCfZyXOUbyO** – In order to read the file you can use SSH to execute commands too. `ssh bandit18@bandit.labs.overthewire.org -p 2220 cat readme` would display the password.

**Level 20** – To gain access to the next level, you should use the `setuid` binary in the homedirectory. Execute it without arguments to find out how to use it. The password for this level can be found in the usual place (`/etc/bandit_pass`), after you have used the `setuid` binary.

**Solution - EeoULMCra2q0dSkYj561DX7s1CpBuOBt** - The file is owned by `bandit20` however, by using `ls -l` you can see `bandit19` is a group with `'-rwsr-x-'` which means `bandit19` can execute the binary as `bandit20`. We need to switch to user `bandit20` to read the file. The command `./bandit20-do` and `cat /etc/bandit_pass/bandit20`